S2 Text. Audit and saved-result reproduction

This appendix records provenance and reproducibility separately from the scientific methods in S1 Text. Computational checks do not add participants or certify that an annotated task window contains continuous motor imagery.

Scientific identities and historical preservation

The CPU baselines, 54 source-only EEGNet fits and single frozen-checkpoint query evaluation are separate stages. The first 18 accepted development fits were preserved when the additional 36 were appended. The combined ledger has 54 unique jobs and SHA256 `013668c8d1b1c5d6061f370 1d495cf7770743a1c6fda8678f5db7ffd50992670`. All three seeds (20261012, 20261013 and 20261014) are retained for each of 18 published participant identities.

The query operation used execution commit `d34703ca5bccf7a5b88a91c2ffb2cba893f7cec8` and contract `fb3a429776e77a4ee8de96c8c7f2a833a4d311cedd8f0a9bcec3d0c6e8df225a`. It used the 54 original checkpoints with fixed model parameters, batch-normalization buffers and source-normalization statistics. Audits bound 54 archives, 1,189 input files and 4,290 query identities. This editorial revision does not repeat a fit, model forward or query operation.

The first CSP diagnostic attempt stopped when the process monitor could not read a procfs interface. Its claim, authorization, exception and private backup remain intact. A separately authorized recovery retained the scientific consumer and source/support-only scope and completed 240 pre-

scribed source-run fold/rotation attempts. Historical batch pauses, same-state resumes, deadlines and failure logs also remain in the repository. Training completion, off-machine export, scientific audit, full private R2 GET verification and recorded acceptance were distinct states.

Operational acknowledgements are not independent human replications or cryptographic signatures. The query acknowledgement records review of the complete receipt, result identities and saved arithmetic; it does not claim a second remote GET or rehash of all 147 original objects. The operator's actual full-GET check remains separate. Journal files exclude live access details, credentials, EEG arrays and learned weights.

Saved-prediction QC lineage

The exploratory QC rules were recorded after the original outcomes were known and before computing the sensitivity summaries. The original 250 μV maximum-channel peak-to-peak flag joins 25,740 saved prediction rows to 4,290 included query identities. The wider 4,320-row query ledger also retains the 30 excluded partial query identities. All 144 reconstructed endpoint values, source-query and five-session target BA for 72 participant-by-method rows, match their original saved values exactly.

The portable inputs include all 7,200 event identities, saved method/seed predictions, the fixed analysis configuration and participant endpoints. Independent source-lineage and arithmetic checks reconciled these with the original CPU and EEGNet text records. No model or signal loading was needed. Counts, labels, inclusion rules and missing-class states remain explicit.

Only ten of the 12 primary participants retain all five unflagged target-session endpoints. The paired QC change therefore compares these same ten participants; it does not interpret a 12-versus-10 comparison as an effect of filtering. Pooled-query summaries are separate endpoints. Missing BA remains undefined rather than being replaced by zero or chance.

Reproducing saved numerical results

S1 Data provides the dictionary, member hashes and commands in `REPRODUCE_SAVED_RESULTS.md`. The saved-prediction script accepts explicit portable input and fresh output paths. Its original public-input replay reproduced all 13 scientific QC CSVs byte-for-byte. Separate implementations checked identity joins, eligibility, seed averaging, session weighting, percentile intervals and leave-one-participant-out values. These checks were AI-assisted and do not substitute for the author's personal review.

The package also contains all 2,871 saved epoch-history rows and 54 fit summaries. The original archive review extracted training text and streamed binary members only for integrity hashing. All 54 JSON histories were complete; 37 empty console logs were not interpreted as evidence of zero execution. Source/target scores, every seed and class-collapsed predictions remain reported.

The four figures use exact saved numerical inputs. Fig 4 retains 180 participant/session contrast cells and all negative values. In this revision, Fig 1's generic point-type legend is clarified as participant points colored by method; its data are unchanged. Fig 4 already used a symmetric RdBu scale, which is retained. The producer writes fresh outputs without refitting a model or changing original bootstrap intervals. TIFF dimensions, 600 dpi, RGB/LZW encoding and hashes were checked. The actual installed face is Liberation Sans; Arial/Times family compliance is not claimed.

The post-review paired tests are separately identified in S1 Text, S11. They read saved participant differences, enumerate fixed sign assignments and report all four unadjusted and Holm-adjusted results. Their rules were written after review and outcome exposure; they do not retroactively become a preregistered analysis. Original intervals and results remain unchanged. The complete audit includes numerical boundary diagnostics and the preserved initial assertion failure; no failed scientific attempt or historical result was erased.

Marker and literature search scope

The marker search checked all 360 derivative event tables, the hash-matched saved EDF-annotation audit, source/conversion manifests, the original codebook and task-extraction function, and the complete public author-repository tree. It establishes missing phase records in the retained derivative stream, not missing original public data. The manifest lists 360 original bar-task GDFs; their embedded event arrays and mapping to this derivative remain unparsed.

Official original-study XML was obtained in this revision. Relevant Methods, GR/PAR descriptions, results and correction were read; this does not assert exhaustive reading of every supplement. Vidaurre's LDA-adaptation paper was checked through its complete abstract and bibliographic metadata. Relevant full-text methods of Arvaneh's data-space adaptation study were read. The reading-scope record distinguishes these levels rather than treating a DOI lookup as full-text verification.

Public availability and licensing

The source EEG remains at the versioned NEMAR release cited in the main article. Saved outputs and code are publicly accessible in the repository. The tag `paper2-plos-source-20261016` identifies the approved scientific snapshot at content commit `f2901decfc39b2adecae13253ee4b1159a4f9a4b`. The ethics-statement revision is designated `paper2-plos-ethics-20261016`; its accompanying release manifest records the exact file identities. Private R2 preserves original experiment packages and audit evidence but is not the public data endpoint. The source DOI is supplied; no new archive DOI or completed journal submission is asserted.

The author explicitly approved MIT for original project software and CC BY 4.0 for original manuscript, figures and derived content. `LICENSING.md` and the complete license texts define this scoped grant. Source metadata keep source attribution and terms; third-party software, fonts,

reading materials and CSL styles retain their own licenses. The new package does not bundle the complete environment, CUDA libraries, source-paper text or unrelated datasets.

The author confirmed personal reading and checking and approved the scientific manuscript content on 16 October 2026. That approval concerns content commit `f2901decfc39b2adecae13253ee4b1159a4f9a4b`; the confirmation record is preserved at commit `1a7e1ab8b6253db52544ba185abb33a6bcb4937b`. It does not establish personal inspection of every byte of the new administrative exports or native Word rendering. The existing full AI-assistance disclosure remains unchanged.

According to the author, the research administration office at Chongqing Medical University confirmed on 16 October 2026 an exemption from additional ethical review for this Kumar2024/NEMAR secondary analysis under Article 32, item (1), of the 2023 Chinese ethics-review measures [1]. The author reported no identifier. The office name is a descriptive translation, not a verified official English name, and no institutional document has been received for this record. Article 32 provides conditional permission for exemption; public availability or participant coding alone is not proof of non-sensitive information or irreversible anonymization. The confirmation is not a new IRB approval or a consent waiver, and its date is not represented as approval before the analysis.

This version changes only administrative ethics/author-status declarations and adds the legal reference. The 69-member S1 Data package, scientific files, tables, parameters, predictions, negative outcomes and statistical definitions remain unchanged. No analysis, model fit or query evaluation was added. Full earlier manifests, accepted ledgers, failure evidence and the sealed original documents remain preserved.

1. National Health Commission of the People's Republic of China, Ministry of Education of the People's Republic of China, Ministry of Science and Technology of the People's Republic of China, National Administration of Traditional Chinese Medicine. Measures for the Ethical Review of Life Science and Medical Research Involving Humans. National Health Commission of the People's Republic of China; 2023. Available: https://www.gov.cn/zhengce/zhengceku/2023-02/28/content_5743658.htm